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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF LAND SURVEYING AND GEO-INFORMATICS

Programme	**:**	34478 BSc (Hons) in Land Surveying and Geo-Informatics 34379 HD in Land Surveying and Geo-Informatics			
Subject Title	**:**	Geospatial Database Management and Design			
Subject Code	**:**	LSGI3245	**Session**	**:**	Semester 1, 2022/23
Date	**:**	12 December 2022	**Time**	**:**	12:30 – 14:30
Time Allowed	**:**	2 hours	**Subject Examiner(s)**	**:**	Dr. CAO Rui (LSGI)

This question paper has a total of 6 pages (attachments included).
(Some pages may be intentionally omitted.)

Instructions to Candidates : This paper has **SEVEN** questions.

Please answer **ALL** questions from Question 1 to Question 5.

For Question 6 and Question 7, please choose **ONLY ONE** to answer.

Total marks = 100.

Available from Invigilator : Nil

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DO NOT TURN OVER THE PAGE UNTIL YOU ARE TOLD TO DO SO

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Section A: Answer **ALL** questions.

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Question 3

(a) Please create an example (e.g., tables with columns/rows) to explain the concepts and properties of the primary key and foreign key in the relational model. (10 marks)

(b) What is the general design process of a relational database? Please list the general steps and use 1-2 sentences to describe each step. (10 marks)

Question 4

What are the advantages of storing spatial data in a spatial DBMS as opposed to a traditional file-based system? Name at least FOUR advantages and use 1-2 sentences to explain each advantage that you list. (10 marks)

…/4

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Question 5

Rivers can supply water to cities that are within a certain distance. Suppose we have three tables as follows. The ***cities*** table and ***rivers*** table store the cities as points and rivers as linestrings, respectively; and their spatial relationship is shown in Fig 1. The ***census*** table provides the urban population and area data based on census blocks within cities. The unit of measure is *meter* in the spatial database.

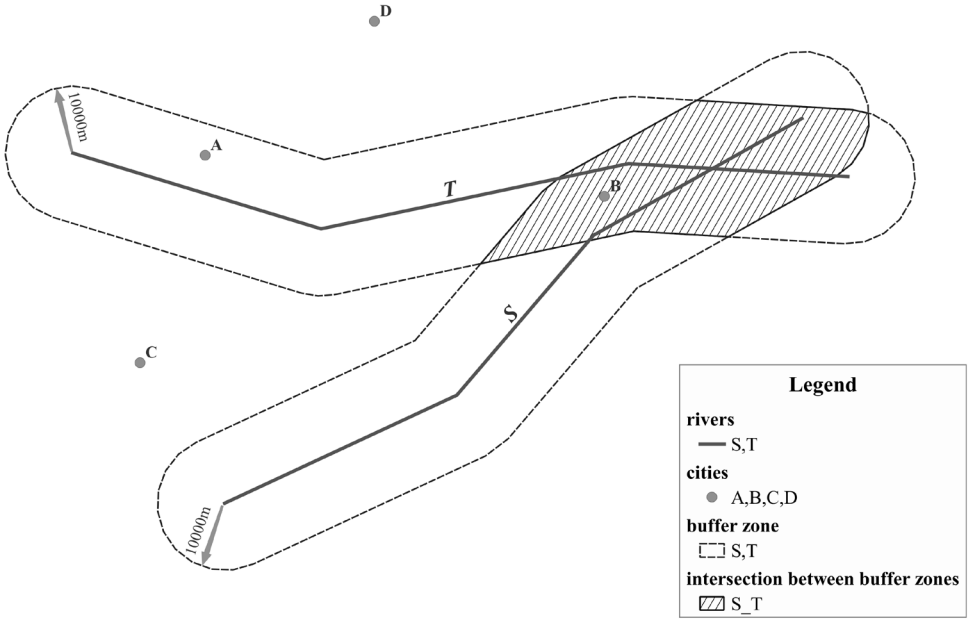


Fig 1: The map of the rivers and the cities.

Table 1: **cities**

name (char)	**geom (geometry(POINT))**
A	POINT(540434.0140486368 3556838.512039512)
B	POINT(599642.3670197944 3550724.6060261857)
C	POINT(530780.478238122 3525947.1974458643)
D	POINT(565533.2071559753 3576789.1527145756)

Table 2: **rivers**

name (char)	**geom (geometry(LINESTRING))**
S	LINESTRING(543330.0747917913 3505031.203189749,577761.0191826274 3521120.429540607,598033.4443847085 3544932.4845398767,628924.758978356 3562308.8489988036)
T	LINESTRING(520805.15790059 3557160.296566529,557620.0862876393 3545878.7295169123,603226.5777777359 555596.850011432,635728.2468424649 3553651.8028787696)

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Table 3: **census** (population: 10^4, area: km^2)

block_id (int)	**city (char)**	**population (numeric)**	**area (numeric)**
1	A	50	28
2	A	23	57
3	B	82	18
4	B	35	42
5	B	42	38
6	C	65	79
8	C	74	63

Based on these tables, we can perform spatial and non-spatial queries to find the information we want. For example, for the SQL expression "SELECT name FROM cities", we can retrieve the following resulting relation:

name
A
B
C
D

Please write down your answer in the form of a relation similar to the above example, for the following SQL expressions: (Note: ***ST_DWithin***(geometry *G1*, geometry *G2*, numeric *D*) returns True if *G1* is within the buffer zone of *G2* with radius of the given distance *D*.)

(a)
```
SELECT * FROM census
WHERE population <= 50 AND area > 40
ORDER BY population DESC;
```
(5 marks)

(b)
```
SELECT name, SUM(area) AS total_area
FROM cities JOIN census ON name = city
GROUP BY name ORDER BY total_area DESC;
```
(5 marks)

(c)
```
SELECT rivers.name AS river_name, cities.name AS city_name
FROM rivers JOIN cities ON ST_DWithin(cities.geom, rivers.geom, 10000)
ORDER BY rivers.name, cities.name;
```
(5 marks)

(d)
```
SELECT rivers.name, SUM(population) AS impact_population
FROM rivers JOIN cities ON ST_DWithin(cities.geom, rivers.geom, 10000)
JOIN census ON cities.name = city
GROUP BY rivers.name ORDER BY rivers.name;
```
(5 marks)

- END OF SECTION A –

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Section B: Answer any **ONE** question.

Question 6

Explain how to use bounding boxes to speed up the computation of intersection between polygons in a spatial database. Please give an example to illustrate. (10 marks)

Question 7

Explain what role space-filling curves (e.g., Z-curve) play in spatial indexing. Please give an example to illustrate. (10 marks)

- END OF PAPER -